Title

- **Project Name:** Vectron
- **Tagline:** VECTRON is an AI-powered codebase dependency explorer. Upload any JavaScript or TypeScript repository as a ZIP and instantly get an interactive knowledge graph of every file, function, class and their relationships.

The average developer spends 58% of their time understanding code, not writing it. VECTRON eliminates that.

- **Team Name:** Algoholic
- **Event:** DevCraft: Simon’s Command Line (or your event name)
- **Date**: 06-04-2006
- **Domain: AI**

Problem Statement

VECTRON is an AI-powered codebase dependency explorer. Upload any JavaScript or TypeScript repository as a ZIP and instantly get an interactive knowledge graph of every file, function, class and their relationships.
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Target Audience

- Developers – To quickly understand and navigate complex codebases
- Students & Learners – To visualize and learn code structure easily
- Open Source Contributors – To reduce onboarding time for new projects
- Development Teams – To improve collaboration and code understanding
- Software Architects & QA Engineers – To analyze dependencies and system design
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Existing Solutions

- Dependency visualization tools (e.g., Dependency Cruiser) show file/module relationships
- VS Code extensions provide basic code structure maps
- Static analysis tools analyze dependencies and architecture
- Graph-based tools visualize code but need manual setup

Gaps & Opportunities

- Lack of AI-driven insights – Existing tools show structure but don’t explain code relationships intelligently
- Limited interactivity – Most visualizations are static and not user-friendly
- Poor granularity – Tools focus on file/module level, not functions, classes, or logic flow
- Time-consuming setup – Many tools require configuration and technical effort
- No real-time understanding – Lack of instant analysis when uploading a project

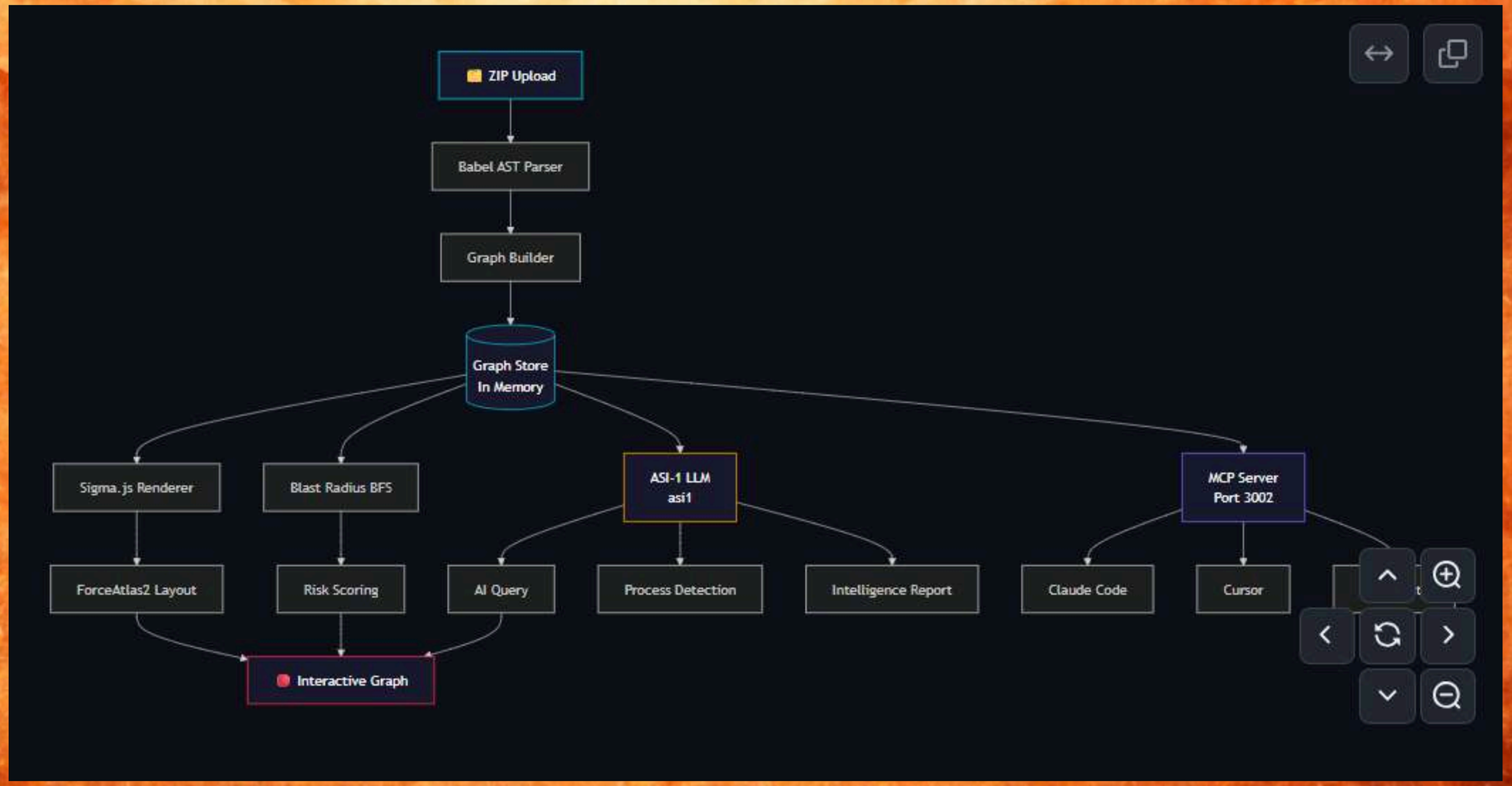
Your Solution

- VECTRON is an AI-powered codebase dependency explorer that transforms complex JavaScript and TypeScript repositories into an interactive knowledge graph. By simply uploading a ZIP file, users can instantly visualize relationships between files, functions, and classes without any manual setup. Unlike traditional tools, VECTRON goes beyond basic dependency mapping by using AI to provide deeper insights into code structure, flow, and interactions. This enables developers, students, and teams to quickly understand unfamiliar codebases, reduce onboarding time, and improve productivity.

Key Features

- AI-Powered Analysis – Automatically understands code structure and relationships
- Interactive Knowledge Graph – Visualizes files, functions, and classes in a dynamic graph
- Granular Insights – Shows detailed connections at file, function, and class level
- Instant ZIP Upload – Analyze entire repositories without setup or configuration
- Dependency Tracking – Easily trace relationships and impact of code changes
- User-Friendly Interface – Simple and intuitive visualization for better understanding
- Faster Onboarding – Helps developers quickly explore and learn new codebases

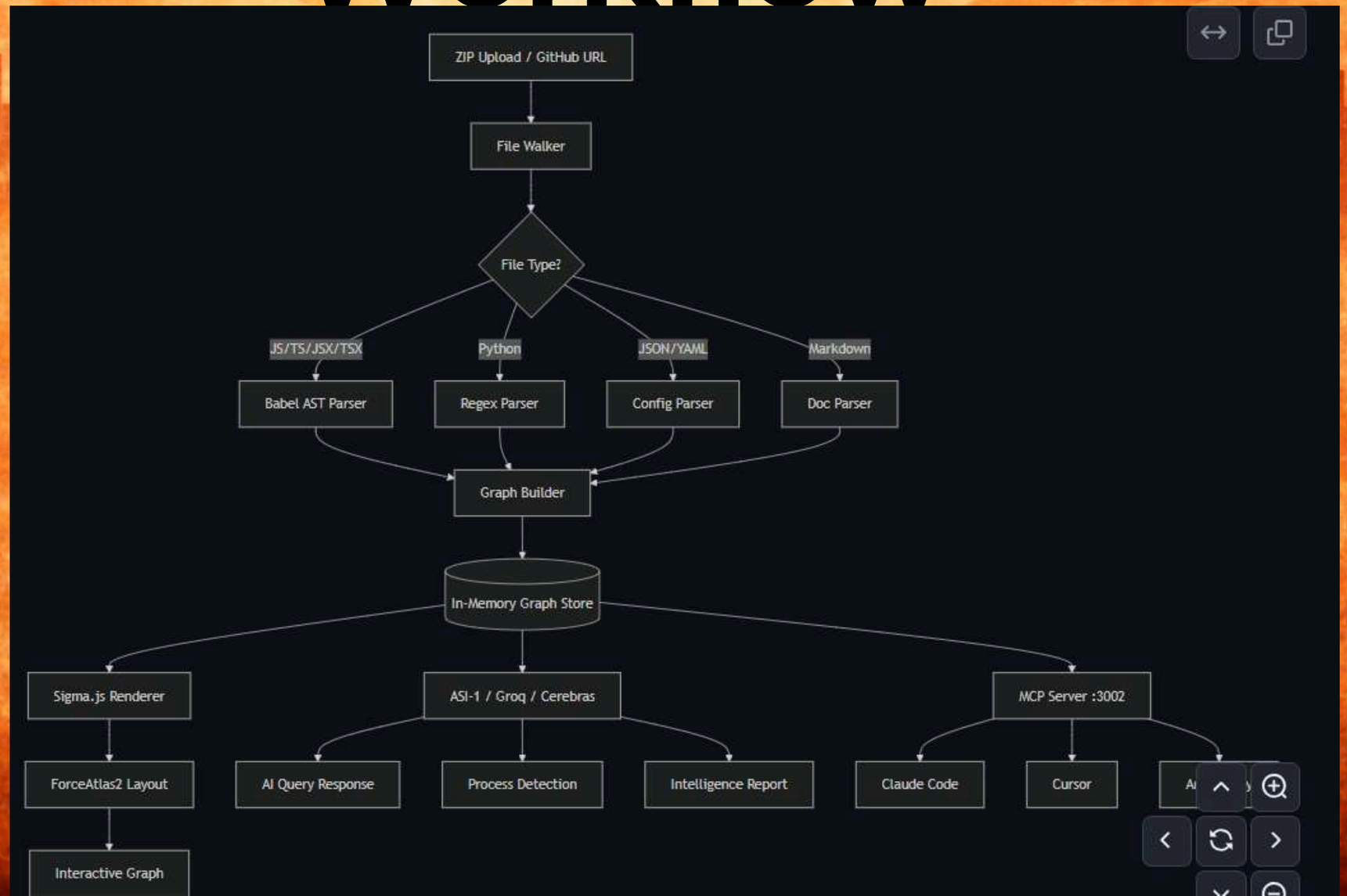
System Architecture



Tech Stack

Layer	Technology
Frontend	React + TypeScript + Vite
Graph Rendering	Sigma.js + Graphology + ForceAtlas2
Backend	Node.js + Express.js
AST Parsing	Babel (JS/TS/JSX/TSX)
AI Primary	ASI-1 (asi1)
AI Fallback	Groq (Llama 3.3 70B), Cerebras (Llama 3.1 8B)
ASI:One API	Multi-agent orchestration platform
Process Diagrams	Mermaid.js
MCP Protocol	@modelcontextprotocol/sdk
Deployment	Railway

Workflow


ZIP Upload / GitHub URL
File Walker
File Type?
JS/TS/JSX/TSX
Python
JSON/YAML
Markdown
Babel AST Parser
Regex Parser
Config Parser
Doc Parser
Graph Builder
In-Memory Graph Store
Sigma.js Renderer
ASI-1 / Groq / Cerebras
MCP Server :3002
ForceAtlas2 Layout
AI Query Response
Process Detection
Intelligence Report
Claude Code
Cursor
Interactive Graph

Impact

Reduces development time by minimizing effort spent understanding complex codebases
Faster onboarding for new developers and contributors
Improves productivity by enabling quick navigation and dependency tracking
Enhances code quality through better understanding of structure and relationships
Supports learning by helping students visualize and grasp real-world codebases
Better collaboration within teams through shared code insights
One-line Summary

Future Scope

Reduces development time by minimizing effort spent understanding complex codebases
Faster onboarding for new developers and contributors
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Enhances code quality through better understanding of structure and relationships
Supports learning by helping students visualize and grasp real-world codebases
Better collaboration within teams through shared code insights
One-line Summary

Team Info

Member 1:
Frontend
Member 2:
Backend
Member 3: api integration
Member 4:
system architect